The General Theory of the Interview

An Information-Theoretic, Game-Theoretic, Psychological, Behavioural-Economic and Ethical Framework

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Abstract

This treatise develops a general interdisciplinary theory of the interview as an information-acquisition, information-allocation, strategic-interaction, investigative and decision-enabling process. The central proposition is that an interview is simultaneously an information-theoretic game of co-operation and competition, conducted between agents possessing heterogeneous information states, preferences, incentives, cognitive capabilities and institutional responsibilities.

The theory integrates information theory, probability, statistics, game theory, decision theory, economics, behavioural economics, psychology, leadership, management, law, ethics, philosophy, political science, diplomacy, international relations, econometrics, computer science, artificial intelligence and cybersecurity. It also considers applications to financial analysis, welfare economics, warfare economics, public administration, scientific investigation and the natural sciences.

An interview is represented as a structured process with a beginning, an investigative phase and an accountable termination. Its participants may cooperate to improve collective understanding while simultaneously competing over disclosure, interpretation, bargaining positions and decision outcomes. Consequently, information acquisition cannot be reduced to maximising the volume of disclosed information.

The framework introduces formal definitions of interview participants, information states, interview groups, interview strategies, disclosure decisions, interview outcomes, reconstruction compatibility and ethical admissibility. It derives a combinatorial representation of interview selection, establishes preliminary information-theoretic results, formulates a constrained interview-allocation problem, and proposes algorithms for selective interviewing.

The treatise further distinguishes interview termination from investigative termination and decision closure. It develops an ethical architecture based on informed participation, confidentiality, voluntary disclosure, legitimate information use, procedural fairness and the protection of affected parties.

The resulting framework is intended to provide a mathematical and institutional foundation for the design of interviews within distributed observation systems, including the Global Systemic Stress Observatory (GSSO). Its governing principle is that the acquisition of information must improve understanding without making the society being observed less free, less secure or less legitimate.

The treatise ends with "The End"

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1 Introduction

An interview is one of the most fundamental mechanisms through which human beings, institutions and organised systems acquire information from one another.

Interviews occur in scientific research, medicine, law, economics, journalism, diplomacy, education, management, intelligence analysis, public administration, political institutions and ordinary social life.

Despite their apparent simplicity, interviews involve several simultaneous processes:

1. Information acquisition.
2. Information disclosure.
3. Strategic interaction.
4. Psychological interpretation.
5. Incentive-dependent behaviour.
6. Leadership and interpersonal coordination.
7. Ethical evaluation.
8. Decision formation.

9. Institutional accountability.

An interview cannot therefore be adequately represented as a unidirectional transmission of information from an interviewee to an interviewer.

Both parties possess information, beliefs, preferences, incentives and limitations. Both may influence the questions, the answers and the interpretation of the resulting information.

The interview is consequently a multi-agent process.

1.1 The central research problem

Let F denote an underlying phenomenon that one or more participants seek to understand.

Let I_i denote the information state of participant i.

In general, participants do not possess identical information:

$$I_i \neq I_j \qquad \text{for some } i, j. \tag{1}$$

This informational asymmetry can produce different interpretations and reconstructions of the same phenomenon.

An interview creates an opportunity to investigate these differences. However, the opportunity does not imply that all information can or should be disclosed.

Participants may have legitimate reasons to withhold information. They may also possess incomplete memories, incorrect beliefs, confidential information, legal obligations or conflicting objectives.

The central problem is therefore:

$$\boxed{\begin{array}{r}\text{How can an interview improve collective understanding} \\ \text{while respecting strategic, psychological, economic} \\ \text{and ethical constraints?}\end{array}} \tag{2}$$

1.2 The central thesis

The interview is simultaneously:

1. An observation mechanism.
2. An information-allocation mechanism.
3. An information-theoretic process.
4. A game-theoretic interaction.
5. A psychological encounter.
6. A behavioural-economic environment.
7. A leadership exercise.
8. An ethical programme.
9. A decision-enabling institution.

The theory developed here seeks to represent these dimensions within a common mathematical architecture.

2 Foundational Concepts and Notation

2.1 The participant set

Let the population of potential participants be

$$P = \{p_1, p_2, \ldots, p_N\}. \tag{3}$$

The population may consist of individuals, institutional representatives, experts, witnesses, researchers, officials or other agents capable of possessing and communicating information.

An agent is defined as an entity capable of holding information, forming beliefs, selecting actions and participating in an interaction.

2.2 Information states

Let $I_i(t)$ denote the information state of participant p_i at time t.

The information state may contain:

- Direct observations.
- Historical records.
- Propositions believed to be true.
- Estimates and forecasts.
- Interpretations.
- Reports received from other agents.
- Information subject to uncertainty.
- Information that the agent is not authorised to disclose.

An information state must distinguish information possessed from information believed, remembered, verified and communicated.

These categories are not necessarily identical.

2.3 Asymmetric information

Definition 2.1. Two agents p_i and p_j possess asymmetric information at time t if their information states differ:

$$I_i(t) \neq I_j(t). \tag{4}$$

The informational asymmetry field is represented by

$$\mathcal{A}(t) = \bigcup_{\substack{i,j=1 \\ i \neq j}}^{N} \left(I_i(t) \setminus I_j(t) \right). \tag{5}$$

This representation follows the basic structure of Asymmetric Informational Ethics developed in the accompanying theoretical programme.

The existence of different information states does not necessarily imply that one participant is better informed in every relevant dimension.

One participant may possess better historical information, another may possess more precise measurements, and a third may possess superior institutional knowledge.

2.4 Information, belief and truth

Let θ represent the true state of the phenomenon.

For each agent i, let

$$\pi_i(\theta) = \Pr_i(\theta \mid I_i) \tag{6}$$

denote the agent's subjective probability distribution over possible states of the phenomenon.

The distinction between information and belief is essential.

A participant may communicate a proposition sincerely while that proposition is false. Conversely, a participant may possess accurate information while lacking sufficient confidence to assert it.

Accordingly,

$$\text{Disclosure} \neq \text{Verification} \neq \text{Truth}. \tag{7}$$

3 The Combinatorial Structure of Interviews

3.1 Interview groups

An interview may involve one or more participants.

For a population of N individuals, the number of distinct interview groups containing exactly n participants is

$$\boxed{\binom{N}{n} = \frac{N!}{n!(N-n)!}} \tag{8}$$

This follows directly from selecting n individuals from N without regard to their ordering.

Proposition 3.1. The number of possible non-empty interview groups drawn from a population of N individuals is

$$\sum_{n=1}^{N} \binom{N}{n} = 2^N - 1. \tag{9}$$

Proof.

By the binomial theorem,

$$(1+1)^N = \sum_{n=0}^{N} \binom{N}{n}. \tag{10}$$

Therefore,

$$2^N = 1 + \sum_{n=1}^{N} \binom{N}{n}. \tag{11}$$

Subtracting the empty-set case gives

$$\sum_{n=1}^{N} \binom{N}{n} = 2^N - 1. \tag{12}$$

□

3.2 The distinction between groups and sequences

The binomial coefficient counts distinct groups.

It does not count interview sequences, question sequences or possible disclosure strategies.

If an interview involves an ordered sequence of n distinct participants selected from N, the number of possible sequences is

$$P(N,n) = \frac{N!}{(N-n)!}. \tag{13}$$

Thus, the number of possible interview arrangements depends on whether participant ordering is relevant.

3.3 Selective interviewing

Exhaustive interviewing is generally neither necessary nor economically desirable.

The combinatorial space may be extremely large even for moderate values of N.

Consequently, the interview system must select a subset of possible interviews.

Let $\mathcal{J}$ denote the set of candidate interviews.

Let x_j indicate whether candidate interview j is selected:

$$x_j \in \{0,1\}. \tag{14}$$

The selection problem is developed in Section 12.

4 The Interview as an Information-Theoretic Process

4.1 Entropy

Let Θ be a random variable representing the unknown state of a phenomenon.

Its Shannon entropy is

$$H(\Theta) = -\sum_{\theta \in \Theta} p(\theta) \log_2 p(\theta). \tag{15}$$

Entropy measures uncertainty under the specified probability distribution.

It does not directly measure the economic importance, ethical importance or practical consequences of uncertainty.

4.2 Conditional entropy

Given information X, the remaining uncertainty about Θ is

$$H(\Theta \mid X). \tag{16}$$

An interview may provide information Y.

The expected reduction in uncertainty is

$$I(\Theta; Y \mid X) = H(\Theta \mid X) - H(\Theta \mid X, Y). \tag{17}$$

This is the conditional mutual information between the unknown phenomenon and the interview result.

Definition 4.1. The expected information contribution of an interview j is

$$\mathcal{I}_j = I(\Theta; Y_j \mid X), \tag{18}$$

where X represents information already available before the interview.

4.3 Information gain and reconstruction quality

Information gain is not identical to reconstruction quality.

An interview may reduce entropy while introducing biased, irrelevant or poorly verified information.

Let $L(\widehat{\theta}, \theta)$ denote a loss function associated with reconstructing the phenomenon.

The expected reconstruction risk before an interview is

$$R_0 = \min_a \mathbb{E}[L(a, \Theta) \mid X]. \tag{19}$$

After receiving interview information Y_j, the expected reconstruction risk becomes

$$R_j = \mathbb{E}_{Y_j \mid X} \left[\min_a \mathbb{E}[L(a, \Theta) \mid X, Y_j] \right]. \tag{20}$$

The expected value of sample information is

$$EVSI_j = R_0 - R_j. \tag{21}$$

Under a correctly specified Bayesian decision model, with the interview information available to the decision-maker and no additional costs, the expected value of sample information is non-negative.

This follows because the decision-maker can always ignore the additional information and retain the original action.

However, the realised outcome of a particular interview need not improve reconstruction.

4.4 Information quality

Let $q_j \in [0, 1]$ denote a provisional reliability parameter associated with an interview result.

This parameter may depend on:

- Source reliability.
- Measurement quality.
- Memory limitations.
- Verification.
- Independence of evidence.
- Incentives to misrepresent.
- Interview conditions.

The parameter q_j must not be interpreted as an objective probability of truth unless an appropriate statistical model justifies that interpretation.

Information quality must be evaluated independently of information quantity.

5 The Interview as a Game of Cooperation and Competition

5.1 Players and strategies

Let an interview involve n participants:

$$\mathcal{P} = \{p_1, p_2, \ldots, p_n\}. \tag{22}$$

Each participant has a strategy set S_i.

Possible strategies include:

1. Full voluntary disclosure.
2. Partial disclosure.
3. Clarification.
4. Verification.
5. Negotiation.
6. Questioning.
7. Non-disclosure.
8. Withdrawal.

A strategy profile is

$$s = (s_1, s_2, \ldots, s_n). \tag{23}$$

Each participant has a utility function

$$U_i(s_i, s_{-i}, \theta), \tag{24}$$

where s_{-i} denotes the strategies of all other participants.

5.2 Cooperative incentives

Participants may benefit from sharing information.

For example, a group of researchers may combine observations to improve the reconstruction of a physical process.

A regulator and a financial institution may exchange information to improve the assessment of systemic risk.

A physician and patient may cooperate to improve diagnosis.

Cooperation may produce a collective surplus:

$$W(s) = \sum_{i=1}^{n} w_i U_i(s), \tag{25}$$

where w_i represents an explicitly specified social or institutional weight.

The weights require normative justification and should not be assumed to be objective.

5.3 Competitive incentives

Participants may also have conflicting interests.

Examples include:

- Bargaining advantages.
- Commercial confidentiality.
- Political competition.
- Reputation management.
- Litigation strategy.
- Institutional self-protection.
- Personal privacy.

The existence of competition does not automatically imply dishonesty.

A participant may legitimately refuse to disclose confidential information.

5.4 The mixed character of interviews

An interview may involve cooperation on one dimension and competition on another.

For example, two negotiating governments may cooperate to establish the facts surrounding a financial dispute while competing over the distribution of its economic consequences.

The interaction is therefore not adequately described by a binary classification of cooperative or competitive behaviour.

Definition 5.1. A mixed interview game is an interview in which the participants possess at least one common informational or investigative interest and at least one potentially conflicting strategic interest.

5.5 Bayesian interview games

Suppose each participant possesses a private type τ_i.

The type may encode private information, preferences, incentives or institutional constraints.

Participants have beliefs over the types of other participants.

A Bayesian strategy is a mapping

$$\sigma_i : \tau_i \longrightarrow \Delta(S_i), \tag{26}$$

where $\Delta(S_i)$ denotes a probability distribution over available strategies.

A Bayesian Nash equilibrium is a strategy profile in which each participant's strategy is optimal given their beliefs about the other participants' strategies and types.

The existence and properties of such an equilibrium depend on the assumptions imposed on the game.

No general theorem guarantees that a particular interview will produce truthful disclosure or socially optimal reconstruction.

6 Psychology and Cognitive Behaviour

6.1 The psychological participant

Participants are not perfect information-processing machines.

Their behaviour is influenced by cognitive capacity, emotion, memory, perception, motivation and social context.

Let Z_i denote the psychological state of participant i.

A simplified representation is

$$Z_i = (M_i, A_i, T_i, C_i, S_i), \tag{27}$$

where:

- M_i: memory state.
- A_i: affective or emotional state.
- T_i: trust.
- C_i: cognitive resources.
- S_i: perceived social safety.

This is a modelling abstraction, not a clinical diagnostic instrument.

6.2 Memory and recall

An interview response may depend on imperfect memory.

Let E_i denote an event experienced by participant i.

The recalled representation is

$$\widehat{E}_i = M(E_i, Z_i, \varepsilon_i), \tag{28}$$

where ε_i represents unobserved recall variation.

Consequently,

$$\widehat{E}_i \neq E_i \tag{29}$$

is possible even in the absence of deliberate deception.

This distinction is particularly important in legal investigations, historical research and scientific interviews.

6.3 Cognitive biases

The interview process may be affected by:

- Confirmation bias.
- Anchoring.
- Availability effects.
- Framing effects.
- Hindsight bias.
- Overconfidence.
- Social desirability.

The interviewer is also susceptible to these biases.

Therefore, an interview methodology must examine both the interviewee and the interviewer.

6.4 Trust and disclosure

Let $T_i(t)$ denote the interviewee's perceived trust in the interviewer.

A provisional behavioural model is

$$D_i(t) = f(T_i(t), S_i(t), C_i(t), K_i(t), R_i(t)), \tag{30}$$

where:

- D_i: disclosure propensity.
- S_i: perceived safety.
- C_i: expected cost of disclosure.
- K_i: perceived confidentiality.
- R_i: perceived relevance of the interview.

This is a proposed empirical specification. Its functional form and parameters require validation.

7 Behavioural Economics and Disclosure Incentives

7.1 Disclosure as an economic decision

A participant may compare the expected benefits and costs of disclosure.

Let:

$$U_i^D = B_i^D - C_i^D \tag{31}$$

denote the utility of disclosure, and

$$U_i^N = B_i^N - C_i^N \tag{32}$$

denote the utility of non-disclosure.

A participant may voluntarily disclose when

$$U_i^D \geq U_i^N. \tag{33}$$

This formulation is deliberately general. It does not imply that all human decisions are consciously calculated.

7.2 Loss aversion and perceived risk

Participants may place greater subjective weight on possible losses than on equivalent gains.

Let v_i represent a reference-dependent valuation function.

The perceived utility of disclosure may then be written as

$$U_i^D = \mathbb{E}[v_i(X_i^D)] - C_i^{\text{privacy}} - C_i^{\text{reputation}}. \tag{34}$$

The relevant costs may include privacy loss, reputational harm, commercial disadvantage and legal exposure.

7.3 Principal-agent problems

An interview may involve a principal seeking information from an agent whose incentives are not perfectly aligned with the principal's objectives.

For example:

- A board interviews management.
- A regulator interviews a financial institution.
- An employer interviews an employee.

- A government interviews a contractor.

The principal may not be able to observe the agent's private information or effort.

This creates problems of adverse selection and moral hazard.

The interview can reduce some information asymmetries but cannot automatically eliminate the underlying incentive problems.

8 Leadership and the Governance of Interviews

8.1 Leadership as an institutional function

Leadership influences the conditions under which information is disclosed, interpreted and acted upon.

The interviewer must exercise authority without converting authority into coercion.

Important leadership functions include:

1. Establishing a legitimate purpose.
2. Clarifying procedural rules.
3. Managing time and participation.
4. Recognising power asymmetries.
5. Protecting minority viewpoints.
6. Preventing intimidation.
7. Distinguishing agreement from acquiescence.
8. Ensuring accountable closure.

8.2 Psychological safety

Psychological safety refers to the perceived ability to participate, ask questions, express uncertainty or report concerns without unreasonable interpersonal punishment.

It can be especially important in organisational interviews.

However, psychological safety does not mean the absence of accountability, scrutiny or disagreement.

8.3 Leadership and information integrity

The interviewer should not manipulate participants into confirming a predetermined hypothesis.

Let Q denote the question-selection process.

A neutral interview design seeks to minimise systematic dependence between the question wording and the desired answer.

In practice, complete neutrality may be unattainable.

The appropriate response is to document assumptions, standardise procedures where suitable, and test for interviewer effects.

9 Top-Down and Bottom-Up Interviewing

9.1 Top-down interviewing

A top-down approach begins with a defined phenomenon or research question.

The sequence is:

1. Define the phenomenon.
2. Identify relevant information deficits.
3. Establish the population of potential participants.
4. Select interview groups.

5. Conduct interviews.
6. Compare evidence.
7. Reconstruct the phenomenon.

Let F denote the target phenomenon and $\mathcal{I}$ the information required to investigate it.
The top-down selection operator is

$$\mathcal{Q}_T : F \longrightarrow \mathcal{I}. \tag{35}$$

This approach is particularly useful when the research question is clearly defined.

9.2 Bottom-up interviewing

A bottom-up approach begins with the population and its distributed information.
The sequence is:

1. Identify participants.
2. Conduct initial interviews.
3. Discover information held by participants.
4. Identify relationships and discrepancies.
5. Discover previously unknown questions.
6. Construct a provisional reconstruction.

The bottom-up construction operator is

$$\mathcal{Q}_B : \mathcal{I} \longrightarrow F^*. \tag{36}$$

Here F^* is a provisional reconstruction rather than an assumption that the complete phenomenon has been discovered.

9.3 Combined interviewing

The two approaches may be combined.

$$F^* \xrightarrow{\mathcal{Q}_T} \mathcal{I} \xrightarrow{\mathcal{Q}_B} F^{**}. \tag{37}$$

The resulting process is iterative.
Top-down investigation tests what the system believes it needs to know.
Bottom-up investigation discovers what the system did not know it needed to know.

10 The Interview Lifecycle

10.1 Beginning, process and end

Every interview has a lifecycle.
Let an interview be represented by

$$\mathcal{J} = (B, I_0, \Phi, D, E), \tag{38}$$

where:

B Beginning, including purpose, scope, participants and consent.

I_0 Initial information states.

Φ Investigative process.

D Decisions or conclusions enabled by the interview.

E End, including closure and information-handling obligations.

10.2 Interview termination

Interview termination occurs when the interview session is formally concluded.

It does not imply that all relevant information has been acquired.

10.3 Investigative termination

Investigative termination occurs when the investigation has reached its specified stopping condition.

The stopping condition may include:

- Adequate evidence.
- Exhaustion of feasible investigative avenues.
- An unacceptable marginal acquisition cost.
- An ethical restriction.
- A defined time limit.
- A decision that further investigation is unnecessary.

10.4 Decision closure

Decision closure occurs when outstanding decisions have been made, deferred under a defined procedure or referred to an appropriate authority.

These three forms of termination need not coincide.

Proposition 10.1. Interview termination does not logically imply investigative termination.

Proof.

An interview may conclude while relevant information remains unresolved.

For example, an interviewee may decline to answer a question, a new issue may be discovered, or further independent evidence may be required.

Therefore, the termination of the interview session does not entail the termination of the investigation. □

10.5 An important ethical qualification

An interview may terminate with unresolved informational asymmetries.

Such an outcome is not necessarily a failure.

An interviewee may legitimately refuse disclosure. Further investigation may be disproportionate, unlawful or harmful.

Consequently, the system must not define successful termination as the compulsory elimination of every information difference.

11 Reconstruction Compatibility

11.1 Different reconstructions

Suppose two participants produce different reconstructions of the same phenomenon:

$$\widehat{F}_1 \neq \widehat{F}_2. \tag{39}$$

The difference may arise from:

- Different observation domains.
- Different temporal references.
- Different measurement precision.
- Different historical information.
- Different model assumptions.

- Different interpretation procedures.
- Genuine structural ambiguity.

Informational asymmetry is a possible explanation, but not necessarily the only explanation.

11.2 Reconstruction operators

Let the reconstruction operator of agent i be

$$\widehat{F}_i = \mathcal{R}_i(I_i, G_i, M_i), \tag{40}$$

where:

I_i Information state.

G_i Observation geometry.

M_i Measurement and reconstruction model.

An interview may improve reconstruction by revealing previously unknown information, assumptions or limitations.

11.3 Compatibility

Let $\mathcal{C}$ denote a reconstruction compatibility operator:

$$\mathcal{C} = \mathcal{C}(\widehat{F}_1, \widehat{F}_2, \mathcal{D}, \Omega, \mathcal{T}), \tag{41}$$

where:

$\mathcal{D}$ Available evidence.

Ω Spatial domain.

$\mathcal{T}$ Temporal domain.

Compatibility should not be confused with agreement.

Two reconstructions may be compatible while describing different aspects of the same phenomenon.

Conversely, two reconstructions may agree while sharing the same underlying measurement or modelling error.

11.4 Preserving ambiguity

Let

$$\mathcal{A}_F = \langle \widehat{F}_1 \mid \widehat{F}_2 \rangle \tag{42}$$

represent an unresolved reconstruction ambiguity.

The system should preserve this ambiguity when the evidence does not justify selecting one reconstruction.

The cost of preserving ambiguity may be represented as

$$C_A = C_{\text{decision}} + C_{\text{delay}} + C_{\text{coordination}} + C_{\text{resolution}}. \tag{43}$$

The cost of premature resolution may be represented as

$$C_P = C_{\text{misclassification}} + C_{\text{irreversibility}} + C_{\text{lost information}}. \tag{44}$$

The appropriate policy must compare these costs while accounting for uncertainty and ethical constraints.

12 Asymmetric Informational Ethics

12.1 Ethics as an investigative process

The interview is not ethically neutral merely because it is described mathematically.

The means of acquiring information matter.

Asymmetric Informational Ethics provides a framework in which ethical programmes are initiated by informational asymmetries, develop through investigation and terminate subject to conditions concerning both residual asymmetries and outstanding decisions.

Let an ethical programme be represented by

$$P_I = \langle E_I, A_I, \Phi_I, D_I \rangle, \tag{45}$$

where:

E_I Legitimate intended end.

A_I Initial asymmetry set.

Φ_I Investigation function.

D_I Decision set.

12.2 Informed participation

Participants should receive sufficient information to understand:

- The purpose of the interview.
- The expected uses of the information.
- The identity and role of the interviewer.
- The relevant confidentiality arrangements.
- The foreseeable risks.
- Their available choices concerning participation.

The form of consent must be appropriate to the setting and applicable legal requirements.

12.3 Voluntary non-disclosure

The right to withhold information is a fundamental consideration.

Information acquisition is not equivalent to unrestricted information extraction.

$$\boxed{\text{Information Acquisition} \neq \text{Unrestricted Information Extraction}} \tag{46}$$

A participant may have legitimate reasons not to disclose information.

Examples include privacy, professional confidentiality, commercial secrecy, legal privilege, personal safety and national security.

The observatory must distinguish information that is available, information that may ethically be requested, information that may lawfully be acquired and information that may legitimately be disclosed to other agents.

12.4 Confidentiality and purpose limitation

Information acquired for one purpose should not automatically be reused for an unrelated purpose.

Let $u(I)$ denote the authorised use of information I.

A purpose-limitation constraint can be represented as

$$\text{Use}(I) \in \mathcal{U}_{\text{authorised}}(I). \tag{47}$$

This is a formal abstraction of a governance requirement, not a substitute for applicable data-protection law.

12.5 Three-component legitimacy

A legitimate interview objective should satisfy three conditions.

1. Compliance with inviolable constraints.
2. Consistency with the legitimate values and commitments of the initiating agent.
3. Defensibility to affected parties under appropriate conditions of disclosure and participation.

These conditions are related to the three-component value architecture developed in Asymmetric Informational Ethics.

The precise content of inviolable constraints requires independent legal and normative specification.

12.6 Ethical admissibility

Let E_j denote the ethical admissibility of interview j.

Define

$$E_j \in \{0, 1\}. \tag{48}$$

An interview that violates a binding ethical constraint receives

$$E_j = 0. \tag{49}$$

The interview must not become admissible merely because its expected information gain is high.

13 Law, Confidentiality and Institutional Accountability

13.1 Legal context

Interviews may occur in different legal environments.

The relevant legal rules may concern:

- Consent.
- Privacy.
- Data protection.
- Professional secrecy.
- Legal privilege.
- Employment rights.
- Evidentiary standards.
- Record retention.
- National security.
- Cross-border data transfers.

The legal status of an interview depends on its jurisdiction, purpose, participants and institutional setting.

No universal interview protocol can replace jurisdiction-specific legal review.

13.2 Procedural fairness

A fair interview procedure should make the process sufficiently transparent to permit scrutiny.

Relevant safeguards include:

1. Clear scope.
2. Consistent question procedures where appropriate.
3. Accurate record-keeping.
4. Opportunities for clarification.
5. Protection against coercion.
6. Appropriate correction mechanisms.
7. Review of consequential findings.

13.3 Evidence and testimony

An interview response is evidence of what was communicated. It is not automatically proof that the communicated proposition is true.

The distinction between testimony, corroboration, documentary evidence and independently verified observation must be preserved.

14 Economics, Welfare and Financial Planning

14.1 Interview costs

Let the total cost of interview j be

$$C_j = C_j^{\text{time}} + C_j^{\text{labour}} + C_j^{\text{technology}} + C_j^{\text{security}} + C_j^{\text{legal}} + C_j^{\text{opportunity}}. \tag{50}$$

The cost of interviewing may also include participant burdens, privacy risks and institutional disruption.

These costs should not be ignored merely because they are difficult to monetise.

14.2 Net value of information

A preliminary net-value expression is

$$NVOI_j = EVSI_j - C_j. \tag{51}$$

This expression is valid only when the benefits and costs are expressed in commensurable units.

Where ethical restrictions are binding, positive net value does not establish admissibility.

14.3 Welfare economics

The social value of an interview may differ from its private value to the interviewer.

Let W denote social welfare.

A proposed welfare function is

$$W = \sum_{i=1}^{N} w_i U_i - C_{\text{system}} - C_{\text{externality}}. \tag{52}$$

The weights w_i must be justified.

The function must also account for privacy, distributional effects, dignity and other values that cannot always be represented adequately by monetary measures.

14.4 Distributional effects

An interview system may impose unequal burdens.

For example, a small group of individuals may be repeatedly interviewed because they are easier to access.

The resulting information system may overrepresent their experiences while underrepresenting less accessible populations.

Therefore, interview allocation should be assessed for distributional bias.

14.5 Financial planning

An interview programme requires a budget.

Let B denote the total authorised budget.

The budget constraint is

$$\sum_{j \in \mathcal{J}} x_j C_j \leq B. \tag{53}$$

A financial plan should distinguish:

- Fixed programme costs.
- Variable interview costs.
- Training costs.
- Security and data-protection costs.
- Independent verification costs.
- Contingency reserves.
- Long-term retention and deletion costs.

14.6 Real-options interpretation

An interview may create the option to conduct further investigation.

The decision to conduct an initial interview may therefore have value beyond the information immediately acquired.

Let V_{followup} denote the expected value of future investigative options.

Then a broader valuation may be represented as

$$V_j = EVSI_j + V_{\text{followup}} - C_j. \tag{54}$$

The value of follow-up options must be estimated without double-counting benefits already included in EVSI.

15 Econometrics and Statistical Validation

15.1 The measurement problem

Interview responses are measured variables.

They may contain systematic and random errors.

Let Y_i denote the observed response of participant i.

A simple measurement model is

$$Y_i = X_i + \varepsilon_i, \tag{55}$$

where X_i is the underlying response construct and ε_i is measurement error.

This model is illustrative; categorical, ordinal, censored and textual responses require appropriate alternatives.

15.2 Interviewer effects

Let J_i denote an interviewer characteristic.

A possible regression specification is

$$Y_{ij} = \alpha + \beta X_i + \gamma J_j + \delta Z_i + u_{ij}, \tag{56}$$

where Z_i contains observed participant characteristics.

The coefficient γ may capture interviewer-associated variation, subject to the identification assumptions of the model.

15.3 Selection bias

Interview participation may be non-random.

Let $R_i = 1$ indicate participation.

If

$$\Pr(R_i = 1 \mid X_i, Z_i) \tag{57}$$

depends on unobserved characteristics associated with the outcome, estimates based only on participants may be biased.

Possible methods include:

- Probability sampling.
- Stratification.
- Post-stratification.
- Inverse-probability weighting.
- Sensitivity analysis.
- Appropriate missing-data models.

No method eliminates selection bias without assumptions.

15.4 Reliability and validity

Reliability concerns the consistency of a measurement procedure.

Validity concerns whether the procedure measures the intended construct.

A highly reliable interview procedure may still be systematically invalid.

The two concepts must be evaluated separately.

15.5 Qualitative and quantitative evidence

Interviews may produce numerical, categorical, narrative or documentary information.

A general framework should permit mixed-method analysis.

Quantitative coding should not erase meaningful contextual information from qualitative evidence.

16 Applications to the Natural Sciences

16.1 Physics

In physics, interviews may involve experimental researchers, instrument operators, engineers and observers.

Their accounts may help identify:

- Instrument calibration errors.
- Experimental boundary conditions.
- Unrecorded interventions.

- Measurement limitations.
- Differences between theoretical and experimental interpretations.

An interview does not replace physical measurement.
It may, however, reveal information necessary to interpret measurement records correctly.

16.2 Chemistry

In chemistry, interviews may assist in documenting experimental procedures, reagent handling, laboratory conditions, process deviations and quality-control practices.

The interview record should not be confused with direct chemical analysis or independent verification.

16.3 Biology

In biology and ecology, interviews may provide information about field observations, local ecological change, experimental history and human interactions with biological systems.

Indigenous and local knowledge may be valuable, but its acquisition requires appropriate consent, attribution and respect for community rights.

16.4 Scientific reproducibility

The interview may reveal undocumented experimental choices.

However, reproducibility ultimately requires sufficiently complete methods, data provenance, measurement records and independent validation.

17 Linguistics and Communication

17.1 Language as a measurement channel

An interview is conducted through language or another communication system.

The meaning received by the interviewer may differ from the meaning intended by the interviewee.

Let M_i denote the intended message and Y_i the interpreted message.

A communication model is

$$Y_i = \mathcal{L}(M_i, \eta_i), \tag{58}$$

where η_i represents linguistic and contextual effects.

These may include:

- Ambiguity.
- Translation.
- Idiom.
- Cultural context.
- Technical vocabulary.
- Pragmatic implication.
- Differences in register.

17.2 Translation and semantic preservation

A translated interview should preserve meaning rather than merely reproduce literal word correspondences.

Where uncertainty remains, the original statement and the translation should be retained separately, subject to applicable confidentiality constraints.

17.3 Question design

Question wording affects responses.

Leading questions, presuppositions and ambiguous terminology may introduce systematic measurement error.

The interviewer should distinguish open-ended, closed-ended, clarifying and verification questions.

18 Psychiatry, Medicine and Clinical Interviews

18.1 Clinical context

Clinical interviews have specific professional and ethical requirements.

They may involve:

- Symptoms.
- Medical history.
- Medication.
- Functional limitations.
- Patient preferences.
- Risk assessment.

Clinical interviews must be conducted within appropriate professional competence and legal frameworks.

18.2 Distinguishing testimony from diagnosis

An interview response is not itself a psychiatric diagnosis.

Clinical conclusions require appropriate assessment, professional judgement and consideration of differential diagnoses.

18.3 Vulnerability and power

Patients may be vulnerable because of illness, distress, dependence or unequal access to medical knowledge.

The interviewer must therefore account for power asymmetry, privacy and the possibility of coercion.

The general theory does not authorise unqualified individuals or automated systems to make clinical determinations.

19 Philosophy, Religion and Moral Reasoning

19.1 Epistemology

The interview is an epistemic process.

It concerns how knowledge is acquired, justified, communicated and challenged.

Different participants may possess different forms of knowledge, including empirical, practical, historical and interpretative knowledge.

19.2 Moral philosophy

An interview may reveal disagreements about values rather than merely disagreements about facts.

The distinction is essential.

A factual disagreement may sometimes be reduced by additional evidence.

A normative disagreement may persist even when the relevant facts are shared.

19.3 Religion and cultural understanding

Religious and cultural interviews may involve deeply held beliefs, identities and community practices.

The interviewer must avoid treating a participant as a representative of every member of a religious or cultural community.

Respectful interviewing requires contextual understanding and protection against stereotyping.

19.4 The limits of consensus

Consensus is not necessarily proof of truth or legitimacy.

A group may agree because of shared error, social pressure or unequal authority.

Therefore,

$$\text{Consensus} \neq \text{Truth}. \tag{59}$$

20 Political Science, Diplomacy and International Relations

20.1 Institutional interviews

Political and institutional interviews may involve multiple levels of authority and accountability.

Participants may represent:

- Governments.
- Ministries.
- Legislatures.
- Regulators.
- Political organisations.
- International institutions.
- Civil society.

The interviewer must distinguish personal views from institutional positions.

20.2 Diplomatic interviews

Diplomatic interviews are often conducted under conditions of strategic ambiguity.

Participants may communicate signals, preserve bargaining space or avoid commitments.

A diplomatic interview may therefore have several simultaneous functions:

1. Information exchange.
2. Relationship maintenance.
3. Negotiation.
4. Signalling.
5. Conflict prevention.
6. Clarification of institutional positions.

20.3 Geopolitical constraints

Information disclosure may have security implications.

The legitimate protection of classified information must be distinguished from illegitimate concealment or misrepresentation.

The general theory does not presume that all information held by a government should be publicly disclosed.

20.4 International cooperation

An interview architecture may help identify information asymmetries between countries.

However, differences in legal systems, institutional mandates, language, data sovereignty and security obligations must be respected.

21 Welfare Economics, Development and Warfare Economics

21.1 Developmental applications

Interviews can reveal information about:

- Infrastructure constraints.
- Labour-market conditions.
- Public-service delivery.
- Poverty and deprivation.
- Financial access.
- Institutional capacity.
- Local development priorities.

The interview system should not assume that GDP or income alone captures welfare.

21.2 Public investment

Interviews may help identify information relevant to the evaluation of infrastructure projects.

However, interview evidence should be combined with engineering, financial, environmental and economic analysis.

21.3 Warfare economics

In national-security contexts, interviews may help assess logistics, institutional preparedness, supply-chain resilience, civilian protection and resource constraints.

Such interviews require strict legal and ethical safeguards.

The objective of a resilience-oriented interview system should be to improve legitimate preparedness and protect human welfare.

It should not be treated as unrestricted intelligence extraction.

21.4 Development-defense trade-offs

The Development–Defense Switching Framework distinguishes developmental, crisis, defense and reconstruction priorities.

An interview system may provide evidence relevant to these different regimes.

Its function is informational.

It should not independently determine national policy.

22 Cybersecurity and Information Governance

22.1 Threat model

Interview records may contain sensitive personal, commercial, financial or institutional information.

Threats include:

- Unauthorised access.
- Data exfiltration.

- Identity disclosure.
- Record tampering.
- Impersonation.
- Phishing.
- Insider misuse.
- Insecure transmission.

22.2 Security objectives

The information architecture should consider:

1. Confidentiality.
2. Integrity.
3. Availability.
4. Authentication.
5. Authorisation.
6. Accountability.
7. Data minimisation.
8. Retention limitation.

22.3 Access control

Let u denote a user and r a record.

An access-control function is

$$A(u, r) = \begin{cases} 1, & \text{if access is authorised,} \\ 0, & \text{otherwise.} \end{cases} \tag{60}$$

Access should be determined by legitimate purpose and authorisation, not merely by technical availability.

22.4 Data provenance

Each record should retain appropriate provenance information:

- Source identifier.
- Time of acquisition.
- Interview method.
- Relevant consent status.
- Verification status.
- Transformation history.
- Access history.

Provenance improves auditability but must itself be protected against misuse.

23 Artificial Intelligence and Machine Learning

23.1 AI-assisted interviewing

AI systems may assist with:

- Transcription.
- Translation.
- Document organisation.
- Question drafting.
- Information retrieval.
- Contradiction detection.
- Thematic classification.
- Reconstruction comparison.

AI assistance does not eliminate the need for human responsibility.

23.2 Model uncertainty

An AI system may produce incorrect transcriptions, misleading summaries or unsupported interpretations.

Let Y denote the original interview record and $\widehat{Y}$ the AI-generated representation.
The transformation error is

$$\varepsilon_Y = d(Y, \widehat{Y}), \tag{61}$$

where d is an appropriately specified distance or loss function.
The choice of d depends on the task.

23.3 Machine-learning applications

Possible tasks include:

1. Speaker diarisation.
2. Topic classification.
3. Semantic retrieval.
4. Named-entity recognition.
5. Sentiment analysis, with appropriate caution.
6. Contradiction candidate detection.
7. Information-gap identification.

Automated sentiment or deception detection should not be treated as reliable proof of a participant's mental state or honesty.

23.4 Human oversight

The system should retain:

- Access to original records.
- Human review of consequential outputs.
- Correction mechanisms.
- Model and version documentation.
- Uncertainty reporting.
- Appropriate appeal procedures.

24 The Selective Interview Allocation Problem

24.1 Formal statement

Let $\mathcal{J}$ denote the set of candidate interviews.
For each interview j, define:

V_j Expected improvement in reconstruction.

C_j Economic and operational cost.

E_j Ethical admissibility.

R_j Relevance to the reconstruction problem.

x_j Selection indicator.

The simplest selection problem is

$$\max_{x_j} \sum_{j \in \mathcal{J}} x_j V_j \tag{62}$$

subject to

$$\sum_{j \in \mathcal{J}} x_j C_j \leq B, \tag{63}$$

$$x_j \leq E_j, \tag{64}$$

$$x_j \in \{0, 1\}. \tag{65}$$

This is a constrained combinatorial optimisation problem.

24.2 Non-additive information contributions

The additive objective is a simplifying approximation.
Two interviews may provide overlapping information.
A third interview may become valuable only after the first two have revealed a previously unknown issue.
Therefore, a more general objective is

$$\max_{S \subseteq \mathcal{J}} V(S) \tag{66}$$

subject to

$$\sum_{j \in S} C_j \leq B, \tag{67}$$

where $V(S)$ is the expected value of the complete selected interview set.
The function $V(S)$ may be non-additive.
Submodularity is one possible structural assumption, but it must be tested rather than presumed.

24.3 Ethical feasibility

The feasible set is

$$\mathcal{F} = \{S \subseteq \mathcal{J} : C(S) \leq B,\ E(S) = 1\}. \tag{68}$$

The selection problem is therefore

$$S^* \in \arg\max_{S \in \mathcal{F}} V(S). \tag{69}$$

The optimisation is performed only over ethically admissible alternatives.

24.4 Distributional constraints

A system may impose additional constraints to prevent excessive interview burdens on particular groups.

For a group g, let $b_g(S)$ denote the burden imposed by interview set S.

A distributional constraint may be

$$b_g(S) \leq \overline{b}_g. \tag{70}$$

The appropriate burden threshold depends on the institutional and legal context.

25 An Adaptive Interview Algorithm

The interview system should be capable of discovering new information deficits during an investigation.

A proposed algorithm is:

```
INPUT:
Population P
Initial phenomenon F
Interview budget B
Ethical constraints E
Reconstruction model R

INITIALISE:
Candidate interview set J
Information state I
Reconstruction F_hat
Outstanding decisions D

WHILE resources remain and investigation is justified:

1. Identify unresolved information asymmetries.

2. Generate candidate interview groups.

3. Determine the investigative purpose of each interview.

4. Evaluate expected reconstruction value.

5. Estimate economic and operational costs.

6. Evaluate legal and ethical admissibility.

7. Exclude inadmissible interviews.

8. Select an interview subject to resource constraints.

9. Obtain appropriate consent and establish conditions.

10. Conduct the interview.

11. Record information, provenance and limitations.

12. Verify information where feasible.

13. Update participant information states.

14. Update the reconstruction.

15. Identify newly discovered asymmetries.
```

```
16. Evaluate whether further investigation is justified.

17. Review outstanding decisions.

END WHILE

OUTPUT:
Reconstruction
Residual uncertainty
Unresolved asymmetries
Interview records
Ethical and legal audit
Decision status
```

The algorithm is a proposed architecture, not a claim that all interviews can be reduced to a deterministic procedure.

26 A General Interview Architecture

The integrated architecture is represented in Figure 1.

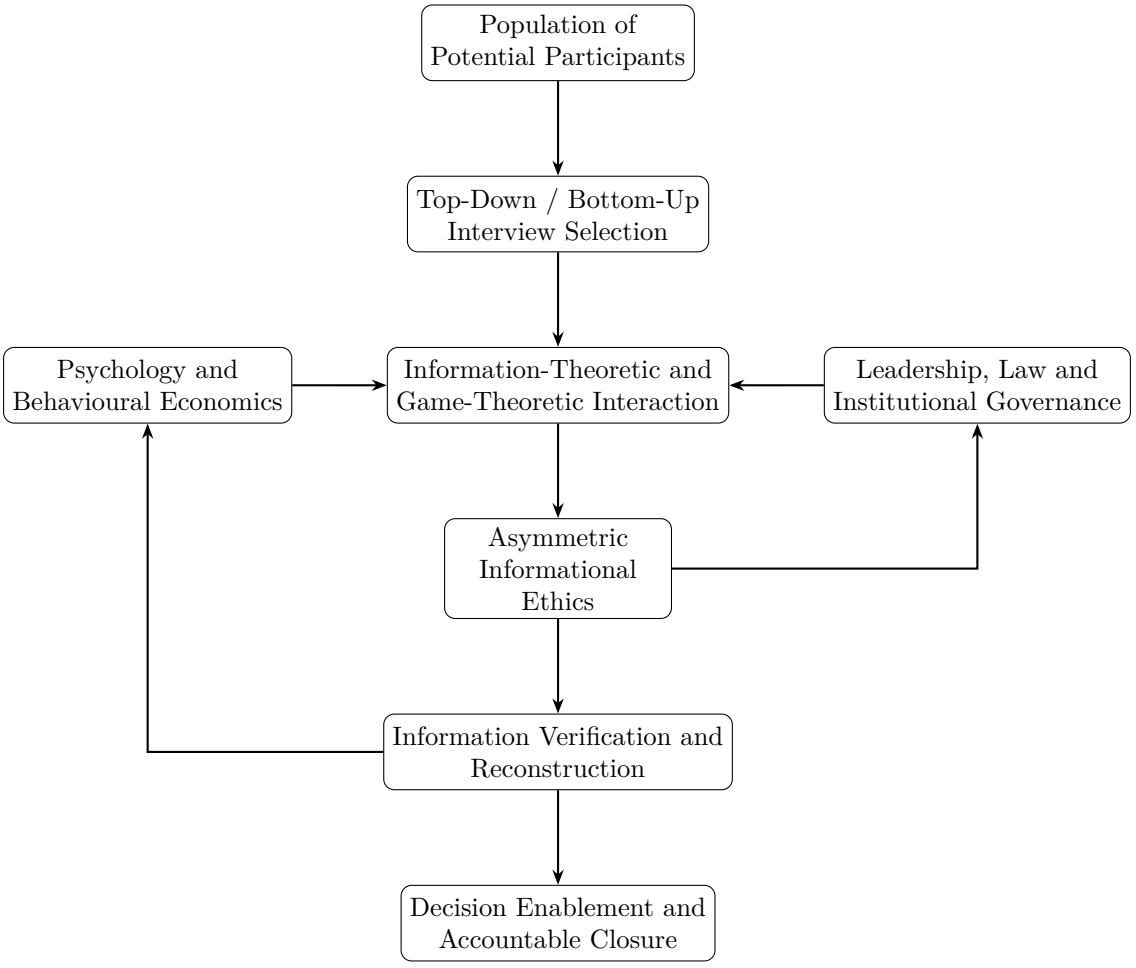


Figure 1: The integrated General Theory of the Interview.

27 Applications to the Global Systemic Stress Observatory

27.1 The GSSO as a distributed observation system

The Global Systemic Stress Observatory is conceived as a distributed architecture involving independent observation agents.

These may include terrestrial, orbital, maritime, aerial and computational agents.

The International Space Station may constitute one independent travelling observation agent within the wider system.

It is not the entire observatory or its necessary central authority.

27.2 The interview layer

The GSSO may use interviews to acquire information from:

- Researchers.
- Financial institutions.
- Public authorities.
- Infrastructure operators.
- Economists.
- Engineers.
- Local communities.
- Independent observation agents.

The interview layer should distinguish direct observation from testimony, interpretation and inference.

27.3 Information provenance

Each observation or interview-derived record should contain appropriate metadata.

A record may be represented as

$$O_j = (s_j, t_j, d_j, m_j, q_j, c_j), \tag{71}$$

where:

s_j Source.

t_j Time.

d_j Domain.

m_j Method.

q_j Quality and uncertainty information.

c_j Consent and access conditions.

27.4 Separation of observation and policy

The GSSO should supply evidence, uncertainty and reconstruction outputs.

It should not independently determine national policy.

A useful institutional separation is:

Observation supplies information.
Reconstruction evaluates evidence.
Economics evaluates consequences.
Institutions make decisions.

The Development–Defense Switching Framework may use the resulting information to inform development, crisis, defense and reconstruction decisions.

28 Empirical Research Programme

28.1 Research questions

The theory generates several testable questions.

1. Does structured interviewing reduce reconstruction error relative to passive information collection?
2. Under what conditions does top-down interviewing outperform bottom-up interviewing?
3. Under what conditions does a hybrid approach outperform either approach individually?
4. How do trust and perceived confidentiality affect voluntary disclosure?
5. How do interviewer characteristics affect response distributions?
6. How does strategic competition affect information reliability?
7. Can ethical constraints improve long-run institutional trust and participation?
8. Does selective interviewing produce better reconstruction per unit of cost than exhaustive interviewing?

28.2 Experimental design

A controlled experiment may compare:

1. Passive observation.
2. Top-down interviews.
3. Bottom-up interviews.
4. Hybrid interviews.
5. Adaptive selective interviews.

The experiments should control for:

- Population composition.
- Interview budget.
- Question design.
- Participant incentives.
- Interviewer training.
- Information availability.
- Evaluation criteria.

28.3 Outcome measures

Potential measures include:

Reconstruction error Difference between estimated and known underlying states in controlled experiments.

Information gain Conditional mutual information.

Decision quality Expected loss under a specified decision problem.

Interview cost Direct and indirect resource use.

Participation Voluntary response and completion rates.

Calibration Agreement between stated probabilities and observed frequencies.

Fairness Distribution of burdens and benefits.

Ethical compliance Documented adherence to applicable safeguards.

28.4 Statistical analysis

For a controlled comparison between two interview designs, let L_i^A and L_i^B denote reconstruction losses under designs A and B.

The paired difference is

$$\Delta_i = L_i^A - L_i^B. \tag{72}$$

The average treatment contrast is

$$\overline{\Delta} = \frac{1}{n} \sum_{i=1}^{n} \Delta_i. \tag{73}$$

Inference should account for clustering, repeated observations, selection effects and dependence between participants where appropriate.

Statistical significance should not be confused with economic importance or ethical legitimacy.

29 Limitations and Open Problems

The proposed theory has several limitations requiring further research.

1. **Information measurement:** Information states are difficult to observe completely.
2. **Strategic complexity:** Real participants may have multiple, changing and partially unknown objectives.
3. **Psychological heterogeneity:** No single behavioural model adequately describes every participant.
4. **Ethical specification:** The content of inviolable constraints requires independent normative and legal justification.
5. **Economic valuation:** Privacy, dignity and trust cannot always be expressed adequately in monetary units.
6. **Causal identification:** Interview effects may be confounded by participant selection and institutional context.
7. **Strategic non-disclosure:** Some information asymmetries may remain unresolved.
8. **Institutional variation:** Legal and cultural differences may prevent universal implementation.
9. **AI limitations:** Automated interpretation may introduce new errors and information asymmetries.
10. **Termination:** An investigation may remain open indefinitely or may appropriately terminate with unresolved questions.

The theory should therefore be developed through formal analysis, controlled experiments, institutional evaluation and external validation.

30 Conclusion

The interview is much more than a question-and-answer session.

It is an information-theoretic process, a strategic game, a psychological encounter, a behavioural-economic interaction, a leadership exercise and an ethical programme.

Its participants may cooperate to improve collective understanding while competing over information, incentives, interpretation and outcomes.

The combinatorial structure determines the possible interview groups, but selective interviewing determines which of those groups should actually be approached.

Information theory explains how uncertainty may be reduced. Game theory explains strategic disclosure and non-disclosure. Psychology explains cognitive and interpersonal mechanisms. Behavioural

economics explains incentives and bounded rationality. Leadership governs the interaction. Law establishes institutional boundaries. Ethics determines which methods and ends are legitimate. Statistics and econometrics evaluate the resulting evidence. Computer science and AI provide tools for organising and analysing information.

The interview also has a beginning, a process and an end.

Its termination need not imply that every information asymmetry has disappeared. Some questions may remain unresolved, some participants may decline disclosure, and some information may remain inaccessible for legitimate reasons.

The purpose of the interview is not to extract the maximum possible amount of information.

It is to acquire information that can legitimately improve understanding and decision-making.

This distinction is fundamental.

$$\boxed{\text{Maximum Information} \neq \text{Optimal Information}} \tag{74}$$

The optimal interview must account for informational value, economic cost, strategic incentives, psychological conditions, legal restrictions and ethical admissibility.

For the Global Systemic Stress Observatory, this framework provides a potential architecture for investigating distributed information asymmetries among independent observation agents, institutions and individuals.

For the wider Council for Planet Earth, it supplies a general method of learning from a complex world without treating that world, or its inhabitants, as objects of unrestricted extraction.

The governing principle is:

$$\boxed{\text{A Better Observatory Must Not Become a Worse Society.}} \tag{75}$$

A Suggested Interview Record

A structured interview record may contain the following fields:

Field	**Description**
Interview identifier	Unique record identifier
Purpose	Investigative objective
Participants	Authorised participant references
Date and time	Interview temporal metadata
Location or medium	Physical or digital setting
Method	Structured, semi-structured or unstructured
Consent status	Applicable participation conditions
Information source	Origin of each material claim
Evidence status	Verified, unverified or disputed
Confidentiality	Access and disclosure restrictions
Outstanding issues	Unresolved questions
Closure status	Open, concluded or referred
Retention status	Applicable record-management rule

Table 1: Suggested interview record.

B Further Research Directions

The following research directions are proposed:

1. A formal theory of interview games with heterogeneous information and preferences.
2. A theory of interview selection under submodular and non-submodular information value.
3. A dynamic model of trust and voluntary disclosure.
4. A theory of strategic ambiguity in diplomatic interviews.

5. A statistical framework for interviewer effects.
6. A formal model of interview termination with residual ambiguity.
7. An economic model of ethical information acquisition.
8. A multi-agent interview protocol for distributed observation systems.
9. An AI-assisted interview architecture with human accountability.
10. An empirical evaluation of top-down, bottom-up and hybrid interview designs.

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Glossary of Principal Definitions

Agent
An entity capable of possessing information and selecting actions.

Information State
The information held by an agent at a particular time.

Asymmetric Information
A condition in which two or more agents possess different information states.

Interview
A structured interaction through which participants communicate, investigate or withhold information.

Interview Group
A subset of the population selected to participate in an interview.

Information Gain
The reduction in uncertainty associated with additional information, under a specified probability model.

Reconstruction
An estimate or representation of an underlying phenomenon.

Reconstruction Compatibility
The degree to which different reconstructions can be reconciled with their respective domains, assumptions and evidence.

Ethical Admissibility
The condition that an interview method or objective satisfies applicable ethical constraints.

Interview Termination
The formal conclusion of an interview session.

Investigative Termination
The conclusion of the investigative process under specified stopping conditions.

Decision Closure
The completion, deferral or referral of outstanding decisions under an appropriate procedure.

Top-Down Interviewing
An approach beginning with a defined phenomenon or research question.

Bottom-Up Interviewing
An approach beginning with participants and their distributed information.

Selective Interviewing
The allocation of limited interview resources to selected candidate interviews.

Mixed Interview Game
An interview involving both cooperative and competitive participant incentives.

The End